Objects and null model

Let $S_t(s)$ be the raw price at time $s \in [0,1]$ on day t.

Let $P_t(s) = 100 \log(S_t(s))$, 100× the log price.

$X_t(s) = P_t(s) - P_t(0)$ is the CIDR curve, giving the percentage return since market opening. Note $X_t(0) = 0$.

$J_t = P_t(0) - P_{t-1}(1)$ is the overnight change in log-price, multiplied by 100, which we call the **overnight jump**.

Each day contributes a pair (J_t, X_t), order is

$$F_{t-1} \to J_t \to X_t.$$

So, the model has to specify two

conditional distributions: $J_t | F_{t-1}$ and $X_t | F_{t-1}, J_t$ (we can recover the new price by summing increments).

Error curve: the CIDR is a cumulative sum of n-Minute Log returns, i.e.

$$X_t(S_k) = \sum_{j \le k} r_{t,j}$$

where $r_{t,j} = X_t(S_j) - X_t(S_{j-1})$. We assume increments are independent, stationary, and continuous. Then the centred CIDR is $\sigma_\varepsilon W_t(S)$, where $W_t(S)$ is a standard Brownian motion. We now have

$$\varepsilon_t(0) = 0, \quad Cov(\varepsilon_t(S), \varepsilon_t(S')) = \sigma_\varepsilon^2 \min(S, S').$$

The null model is therefore

$$J_t = \alpha + \eta_t;$$

$$X_t(s) = \mu(s) + \sigma_\varepsilon W_t(s),$$

where

- $r_t \overset{i.i.d.}{\sim} N(0, \sigma_r^2)$;
- λ is the mean overnight jumps;
- $\mu(s)$ is the mean intraday return;
- σ_ε is the intraday volatility.

This reflects H_0: overnight jumps are not predictable from intraday return data.

Use of logs and setting $X_t(0) = 0$

Percentage volatility does not depend on the price level in equity markets. We would like our model to reflect this.

A simple starting point is geometric Brownian motion, with i.i.d. overnight jumps.

$$dS_\tau = \nu S_\tau d\tau + \sigma S_\tau dB_\tau,$$

with solution

$$S_\tau = S_0 \exp\left(\underbrace{\left(\nu - \frac{\sigma^2}{2}\right)}_{\mu}\tau + \sigma B_\tau\right)$$

Suppose the overnight closure has length $T > 0$. So, day t's trading session is $[a_t, a_t + 1]$, where $a_t = t(1+T) - 1$. The price curve on day t is therefore

$$S_t(s) = S(a_t + s), \quad s \in [0,1].$$

and the moments are

$$E(S_t(s)) = S_0 e^{\nu(a_t + s)}$$

$$\mathrm{Var}(S_t(s)) = S_0^2 e^{2\nu(a_t+s)}\left(e^{\sigma^2(a_t+s)} - 1\right).$$

Since a_t grows linearly in t, the mean and variance of $S_t(s)$ depend on t, so $\{S_t\}$ is not stationary.

What if we take increments? Let

$$Y_t(s) = S_t(s) - S_t(0)$$

$$= S(a_t + s) - S(a_t).$$

This gives

$$E(Y_t(s)) = S_0 e^{\nu(a_t+s)} - S_0 e^{\nu a_t}$$

$$= S_0 e^{\nu a_t}\left(e^{\nu s} - 1\right).$$

The only way this is constant is if $\nu = 0$, but even still, we get

$$\mathrm{Var}(Y_t(s)) = \mathrm{Var}(S_t(s)) + \mathrm{Var}(S_t(0)) - 2\mathrm{Cov}(S_t(s), S_t(0))$$

$$= S_0^2\left(e^{\sigma^2(a_t+s)}-1\right)+S_0^2\left(e^{\sigma^2 a_t}-1\right)-2\mathrm{Cov}(\dots)$$

$$= S_0^2\left(e^{\sigma^2 a_t}\left(e^{\sigma^2 s}+1\right)-2\right)$$

$$-2\left(E\left(S_t(s)\,S_t(0)\right)-\underbrace{E(S_t(s))\,E(S_t(0))}_{S_0^2}\right)$$

$$= \dots$$

$$= S_0^2 e^{\sigma^2 a_t}\left(e^{\sigma^2 s}-1\right),$$

which depends on a_t, so it is still not a stationary process.

If we condition on market open,

$$\mathrm{Var}\left(Y_t(s)\mid S_t(0)\right) = \left(S_t(0)\right)^2\left(e^{\sigma^2 s}-1\right),$$

so we can't realistically model volatility with a simple (additive) model.

Now set $P_T = 100 \log S_T$, yielding

$$P_T = 100 \log\left(S_0 \exp(mT + \sigma B_T)\right)$$

$$= 100 \log(S_0) + 100(mT + \sigma B_T)$$

$$= P_0 + 100(mT + \sigma B_T).$$

This still has a variance which depends on T

$$\mathrm{Var}(P_t(s)) = 10^4 \mathrm{Var}(\sigma B(a_t + s))$$

$$= 10^4 \sigma^2 (a_t + s).$$

The fix is to take increments of Log prices:

$$X_t(s) = P_t(s) - P_t(0)$$

$$= 100(ms + \sigma(B(a_t + s) - B(a_t))).$$

So,

$$E(X_t(s)) = 100ms;$$

$$Var(X_t(s)) = 10^4\sigma^2 s;$$

$$Cov(X_t(s), X_t(s')) = 10^4\sigma^2 \min(s, s').$$

In short:

- Arithmetic difference only, i.e. U_t, leads to exponential growth of variance;
- Log only leads to linear growth of variance;
- Log AND difference leads to i.i.d..

The joint model (J_t, X_t) is

$$X_t(s) = 100ms + \underbrace{100\sigma}_{\sigma_\varepsilon}\left(B(a_t+s) - B(a_t)\right);$$

$$J_t = 100\left(\mu\tau + \sigma\left(B(a_t) - B(a_t - \tau)\right)\right).$$

Every piece is a BM increment over a disjoint interval. So,

- $\{X_t\}$ is i.i.d. Gaussian in $L^2[0,1]$, with:
  - Mean $\mu(s) = 100\mu s$;
  - Covariance $\Sigma(s,s') = 10^4\sigma^2 \min(s,s')$;
  - $X_t(0) = 0$ by default;
- $\{J_t\} \overset{\text{i.i.d}}{\sim} N(\underbrace{100\mu\tau}_{\alpha}, \underbrace{10^4\sigma^2\tau}_{\sigma_n^2})$;
- $\{J_t\}$ is independent of $\{X_t\}$.

So, this GBM is a special case of the real model, and every added term is something we test, i.e. departures from this model are how we test things. We start with our basic proposed model: one which connects the overnight jump to intraday returns.

The basic model

This adds the research hypothesis

$$Y_t = \alpha + \int_0^1 \beta(u) X_{t-1}(u)\, du + \eta_t ,$$

$$X_t(s) = \mu(s) + \sigma_\varepsilon W_t(s).$$

We assume

- $\eta_t \overset{i.i.d}{\sim} N(0, \sigma_\eta^2)$, independent of $\mathcal{F}_{t-1}$;
- $\{W_t\}$ i.i.d standard BMs,
- $\mu(0) = 0$;
- $\{\eta_t\}$ independent of $\{W_t\}$